

UnifiedBus™ (UB) Base Specification

Revision: 2.0

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1 Introduction



1.1 Purpose

This document defines the UnifiedBus (UB) Base Specification Revision 2.0. It specifies the UB architecture, protocol stack, and programming models and shall serve as the normative reference for the design, implementation, verification, and validation of UB-compliant devices and systems. This document also defines the mandatory and optional features, behavioral requirements, interoperability rules, configuration mechanisms, and memory and resource management models that shall be supported by all implementations claiming conformance with the UB Base Specification.

This document is intended for a wide range of readers, including but not limited to:

- Chip and IP designers, developers, and verification and validation engineers
- Firmware and software architects and developers
- System and board-level hardware designers
- Standards and compliance engineers
- Product managers and technical marketing personnel

1.2 Scope

This document describes the UB architecture, physical layer, data link layer, network layer, transport layer, transaction layer, and function layer, as well as other key functions and features to help readers understand and implement UB effectively.

1.3 Organization

The UB specification series consists of the UB Base Specification (this document) and several complementary specifications. The UB Base Specification defines the complete protocol stack, electrical specifications, packet formats, behavioral rules, and conformance requirements that every compliant implementation shall satisfy. The complementary specifications include the UB Firmware Specification, UB Software Reference Design for Operating Systems, and other related specifications that facilitate understanding and product development.

1.4 Specification Conventions

The keywords defined below, when appearing in ALL CAPITALS, signify specific compliance requirements and levels of obligation:

- Mandatory requirements: "SHALL" and "SHALL NOT"
- Optional requirements: "MAY" and "MAY NOT"
- Recommendations: "RECOMMENDED" and "NOT RECOMMENDED"

1.5 Normative References

1. IANA, "Address Resolution Protocol (ARP) Parameters"
2. IANA, "Service Name and Transport Protocol Port Number Registry"
3. IEEE Std 802.3™-2022, "IEEE Standard for Ethernet"
4. IEEE Std 802.3ck™-2022, "IEEE Standard for Ethernet Amendment 4: Physical Layer Specifications and Management Parameters for 100 Gb/s, 200 Gb/s, and 400 Gb/s Electrical Interfaces Based on 100 Gb/s Signaling"
5. IEEE 802.1AX-2020, "IEEE Standard for Local and Metropolitan Area Networks--Link Aggregation"
6. IETF RFC 768, "User Datagram Protocol (UDP)"
7. IETF RFC 791, "Internet Protocol"
8. IETF RFC 8200, "Internet Protocol, Version 6 (IPv6) Specification"
9. IETF RFC 2131, "Dynamic Host Configuration Protocol"
10. IETF RFC 8415, "Dynamic Host Configuration Protocol for IPv6"
11. DMTF DSP0236, "Management Component Transport Protocol (MCTP) Base Specification"
12. DMTF DSP0274, "Security Protocol and Data Model (SPDM) Specification"
13. NIST SP 800-38D, "Recommendation for Block Cipher Modes of Operation: Galois/Counter Mode (GCM) and GMAC"
14. OIF, "CEI-112G-LINEAR-PAM4"

1.6 Terminology

Term	Definition
cell	The basic unit of a credit, used for credit-based flow control. A cell consists of one or more flits. The number of flits per cell is configurable.
compact network address (CNA)	A shortened network address format supporting 16-bit and 24-bit representations.
compact transport (CTP)	A transport layer mode that leverages underlying protocols to provide reliable and congestion-controlled transport services.
configuration space	A device storage area used to hold information such as the device's capabilities, status, and configuration, providing the software with an interface for Entity configuration management.
data link layer data block (DLLDB)	A data unit that constitutes a data link layer packet, with a length of 1 to 32 flits.
Entity	The basic unit by which a device allocates its own resources. Each Entity is a communication object within a UB domain.
Entity identifier (EID)	An identifier assigned to an Entity that uniquely identifies the communication object identity of that Entity within a UB domain.

Term	Definition
execution environment bits (EE_bits)	An identifier bitfield that indicates the security state of the computing environment from which a trusted execution environment (TEE) secure communication request originates.
flit	A fixed length (20 bytes) data link layer unit of transfer and interface with physical layer.
globally unique identifier (GUID)	A globally unique identifier of Entity, assigned at the manufacturing stage.
Home	The memory owner in a Home-User access model.
initiator	An Entity that originates transaction requests.
Jetty	The basic communication unit in unified remote memory access (URMA), which provides the capability to issue and execute asynchronous transaction operations, supporting communication modes such as many-to-many and one-to-one.
link management state machine (LMSM)	The state machine responsible for link training, data rate negotiation, equalization, and fault recovery in the physical layer.
memory segment	A block of continuous virtual/logical addresses that serves as the basic object of memory transaction operations, identified by a globally unique UB memory descriptor (UBMD).
maximum transmission unit (MTU)	The maximum size of the transaction layer payload that a transport layer packet can carry.
network partition	A collection of UB processing units (UBPUs) with assigned IP addresses, where network communication between different network partitions is isolated.
reliable and ordered by initiator (ROI)	A transaction service mode that is reliable and where the transaction ordering is maintained by the initiator, enabling out-of-order transmission of transaction layer packets.
reliable and ordered by lower layer (ROL)	A reliable transaction service mode where transaction ordering is maintained by the lower-layer protocol, enabling out-of-order transmission depending on capabilities of the lower layer.
reliable and ordered by target (ROT)	A transaction service mode that is reliable and where transaction ordering is maintained by the target, enabling out-of-order transmission of transaction layer packets.
reliable transport (RTP)	A transport layer mode that provides end-to-end reliable and duplication-free transmission services.
resource space	A device storage area used to hold interrupt information, device functionality configurations, and vendor-defined information, providing the software with an interface for Entity configuration management.
receive lane (RX lane)	A receive channel of a port. A port typically has 1, 2, 4, or 8 receive lanes. UB supports asymmetric links, meaning the number of receive lanes on a port may differ from the number of transmit lanes.
target	The Entity that receives and processes a transaction request.

Term	Definition
token	A credential used to authenticate whether the request initiator is permitted to access the target memory or Jetty, comprising an identifier (TokenID) or an index representing the receive queue, and a credential value (TokenValue).
transaction completion order (TCO)	The order in which completion notifications are generated after a sequence of transactions is executed. The transaction completion order is categorized into two types: send completion order and receive completion order.
transaction execution order (TEO)	The order in which a sequence of transactions is executed on the target.
transport acknowledgement (TPACK)	A positive acknowledgement packet returned by the transport receiver to the transport sender to indicate successful reception of a transport data packet, used to ensure end-to-end transport reliability.
transport channel (TP channel)	An end-to-end connection established between two transport endpoints. It provides end-to-end reliable communication for the transaction layer.
transport channel group (TPG)	A group of several transport channels that enables load balancing of transaction packets across the channels.
transport endpoint (TPEP)	An endpoint participating in transport layer communication, used for transmitting and receiving transport layer packets.
transport negative acknowledgement (TPNAK)	An error response packet returned by the transport receiver to transport sender, providing explicit error information to ensure end-to-end transport reliability.
transport receiver (TP receiver)	The transport endpoint that receives transport layer packets.
transport sender (TP sender)	The transport endpoint that sends transport layer packets.
trusted execution environment (TEE)	A computing environment built upon hardware-level isolation and a secure boot mechanism to ensure the confidentiality, integrity, authenticity, and non-repudiation of data and code related to security-sensitive applications. [Source: GB/T 41388-2022, 3.3, modified]
transmit lane (TX lane)	A transmit channel of a port. A port typically has 1, 2, 4, or 8 transmit lanes. UB supports asymmetric links, meaning the number of transmit lanes on a port may differ from the number of receive lanes.
UB address (UBA)	The address provided by the home to the user, which is used to access the home's memory segment.
UB decoder	The component of a UB Controller responsible for translating the user physical addresses into UB addresses.
UB domain	A collection of UBPU's interconnected using UB links.
UB Fabric	The collection of all UB Switches and UB links in a UB domain.
UB link	A connection comprising two ports and the TX lanes and RX lanes that connect them. The number of TX lanes and RX lanes in a UB link may differ.

Term	Definition
UB memory descriptor (UBMD)	A UBMD includes the Entity identifier, TokenID, and UB address, used to index the home's physical address.
UB memory management unit (UMMU)	The component that translates a UB memory descriptor into the home's physical address and performs permission validation.
UB over Ethernet (UBoE)	An encapsulation of UB transport layer and upper layers for transporting UB transactions over Ethernet/IP networks, where UB packets are routable over the IP network.
UB partition	A UB partition is a collection of Entities. UB transactional communication is isolated between UB partitions.
UB processing unit (UBPU)	A processing unit that supports the UB protocol stack and implements device specific functions.
unified remote memory access (URMA)	A high-performance asynchronous communication library supporting UB semantics, providing asynchronous memory access and two-sided message communication functions.
unified remote procedure call (URPC)	A remote procedure call protocol, utilizing UB transaction layer capabilities and direct memory access, enabling direct peer-to-peer remote function calls between UBPU's.
UnifiedBus (UB)	An interconnect technology and protocol stack for SuperPoD.
unreliable and non-ordered (UNO)	An unreliable transaction service mode that provides no transaction ordering, allowing out-of-order transmission of transaction layer packets.
unreliable transport (UTP)	A transport layer mode that provides unreliable transport service.
User	The Entity that accesses memory in the Home-User access model.

2 Architecture

2.1 Overview

UB is a high-performance, low-latency interconnect protocol specifically designed for SuperPoD-scale AI and HPC deployments. It ensures seamless communication among diverse processing units and facilitates I/Os and memory access through a single, unified interconnect technology. This design delivers efficient coordination among processing units, high-speed data movement, centralized resource management, flexible resource orchestration, and high-performance programming.

A computing system powered by UB is known as a UB system. A UB system may scale seamlessly from a single server to tens of thousands of processing units while preserving peer-to-peer semantics and dynamic resource pooling. Figure 2-1 and Figure 2-2 illustrate examples of UB systems.

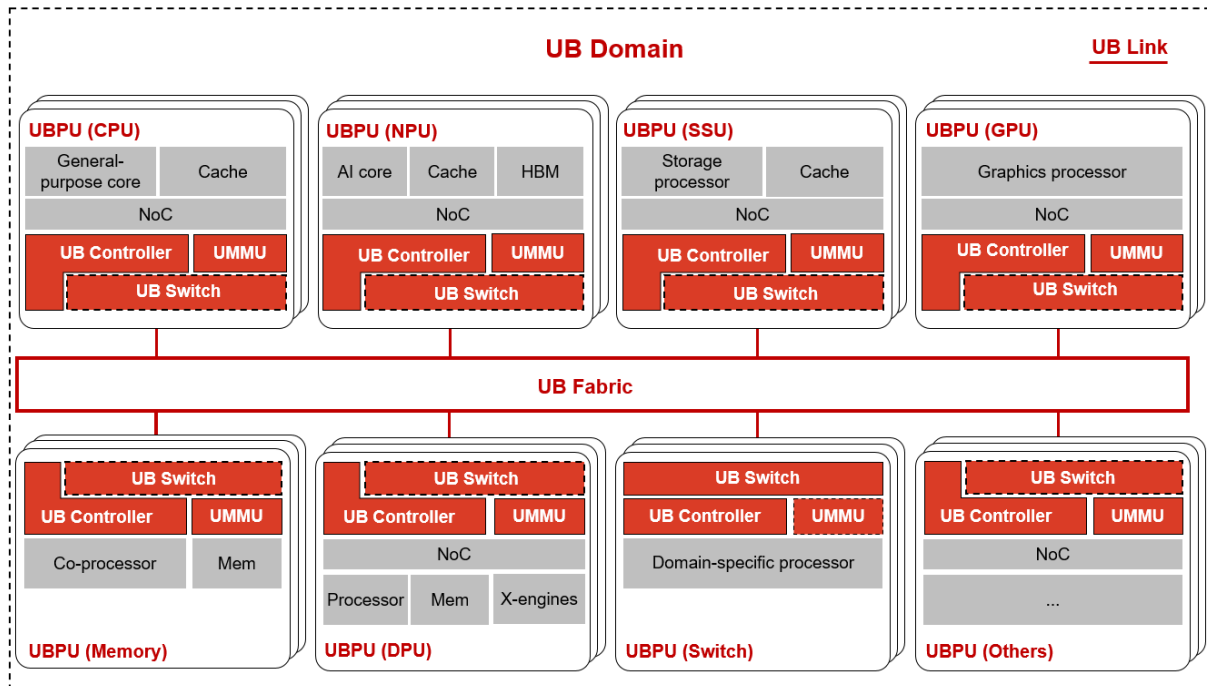


Figure 2-1 UB system within a single UB domain

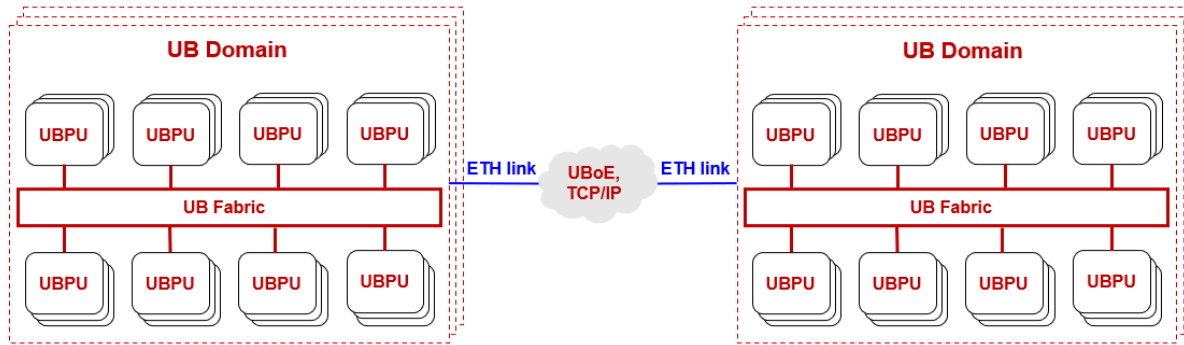


Figure 2-2 UB system across UB domains

A UB system consists of the following fundamental elements:

- A **UB processing unit (UBPU)** is a processing unit that supports the UB protocol stack and implements specific functions.
- A **UB Controller** is a component within a UBPU that implements the UB protocol stack and provides both software and hardware interfaces.
- A **UB memory management unit (UMMU)** is a component within a UBPU responsible for memory address mapping and access permission verification.
- A **UB Switch** is an optional component within a UBPU that forwards packets between UB ports.
- A **UB link** is a full-duplex, point-to-point connection between two UBPU ports.
- A **UB domain** is a collection of UBPUs interconnected via UB links.
- A **UB Fabric** is the collection of all the UB Switches and UB links within a UB domain.
- **UB over Ethernet (UBoE)** is an encapsulation method that transports native UB transaction packets over standard Ethernet/IP networks, enabling interconnection of multiple UB domains.

UB offers the following key features:

- **Unified protocol:** A single protocol handles memory access, messaging, remote procedure calls, and resource management, eliminating protocol conversion overheads and simplifying software development.
- **Peer-to-peer coordination:** Every UBPU is an architectural peer. Any UBPU can directly initiate transactions to any other UBPU and can expose its own functions to or invoke functions from any other UBPU without requiring a host, proxy, or intermediate controller.
- **All-resource pooling:** Each UBPU can share its compute, memory, and storage resources with other UBPU, and at the same time utilize the resources shared by other UBPU within a UB domain, maximizing system resource utilization. UBPU resources can be allocated at the granularity of Entities to different users, allowing flexible orchestration of compute nodes. Compute resource pooling supports elastic scaling of homogeneous compute resources and on-demand orchestration of heterogeneous compute resources for specific tasks. Memory resource pooling enables memory sharing within a UB domain. Interconnect resource pooling optimizes the sharing of diverse interconnect resources, such as transport channels (TP channels) of the transport layer shared across multiple Entities, multi-port aggregation,

and end-to-end multipathing, maximizing bandwidth utilization. Additionally, UB Fabric Managers (UBFMs) are introduced to centrally manage and schedule pooled resources within a UB domain, enabling more organized and efficient resource utilization.

- **Full-stack coordination:** The UB protocol stack employs a layered architecture. Each protocol layer offers multiple selectable modes, enabling the system to be tuned for the exact performance, power, latency, and reliability requirements of each workload.
- **Flexible topology:** UBPU's may integrate switching capability, allowing a wide range of topologies (nD-FullMesh, Clos, torus, etc.) or topology combinations. Virtual lanes, per-packet/per-flow load balancing, TP channel sharing, and adaptive routing provide maximum bandwidth utilization and fault tolerance regardless of the chosen topology.
- **High availability:** UB leverages multiple techniques to ensure high system availability, such as reducing data rates or the number of lanes when a fault occurs and restoring them once it is resolved at the physical layer, point-to-point retransmission at the data link layer, multipathing at the network layer, and end-to-end retransmission at the transport layer. It also supports fault isolation, rapid recovery, and comprehensive Reliability, Availability, and Serviceability (RAS) features at both system and individual layer levels.

2.2 Protocol Stack

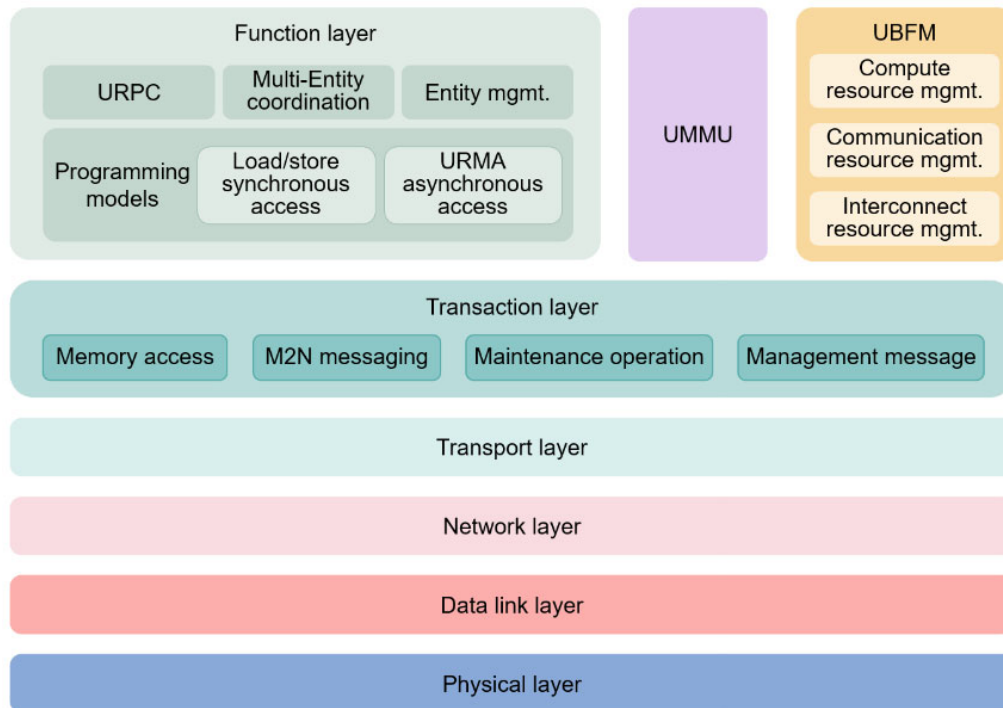


Figure 2-3 UB protocol stack

The UB protocol consists of:

- **Physical layer:** Transmits bit streams over physical media for the data link layer. The physical layer supports customizable data rates to maximize serializer/deserializer (SerDes) and channel performance. It also provides forward error correction (FEC) modes and enables dynamic switching between them to match the bit error rate (BER) characteristics of different links, thereby minimizing latency. The physical layer can reduce data rates or the number of lanes in response to faults and restore them once the faults are resolved, and provide optical module channel protection, enhancing link availability.
- **Data link layer:** Ensures reliable transmission of network layer packets between the two ports of a UB link. The data link layer guarantees error-free data transmission over point-to-point links with cyclic redundancy check (CRC), packet retransmission, credit-based flow control, and virtual lanes.
- **Network layer:** Provides routing services both within and across UB domains for upper-layer protocols. It supports both standard IP addresses and compact network addresses (CNAs), catering to the varied needs of upper-layer protocols for compatibility and efficiency. The network layer also supports multipath communication, per-packet/per-flow load balancing, and customizable routing policies specified by upper layers. These capabilities maximize bandwidth utilization and efficiency in multipath network environments.
- **Transport layer:** Offers multiple end-to-end transport services for the transaction layer. To be specific, reliable transport (RTP) provides a reliable, duplication-free, connection-oriented transport service for lossless end-to-end data transmission. Compact transport (CTP) ensures transport reliability through lower layers such as the data link layer, suitable for communication environments with low end-to-end path failure rates. Unreliable transport (UTP) provides an unreliable, connectionless transport service, typically used in scenarios tolerant of packet loss, such as in-band connection establishment. The transport layer supports unified scheduling of multiple TP channels for load sharing across the channels. It also offers load balancing and congestion management to minimize dynamic transport latency and enhance packet transfer efficiency. Furthermore, the transport layer also supports transport bypass (TP bypass) that allows the transaction layer to directly access network layer services, thereby reducing protocol overheads.
- **Transaction layer:** Offers four types of transaction operations for upper-layer applications (including the functions defined by the UB protocol): memory, messaging, maintenance, and management operations. These operations enable both synchronous and asynchronous memory access, messaging, status maintenance, and Entity management. The transaction layer abstracts the transaction interaction process to hide differences among different programming models. Four transaction service modes are available at this layer: reliable and ordered by initiator (ROI), reliable and ordered by target (ROT), reliable and ordered by lower layer (ROL), as well as unreliable and non-ordered (UNO).
- **Function layer:** Provides two programming models—load/store synchronous access and unified remote memory access (URMA) asynchronous access. In load/store synchronous access, UB Controllers collaborate with the network on chip (NoC) to convert load/store

instructions into transaction operations (read, write, atomic, etc.). In URMA asynchronous access, applications can use APIs provided by Jetties to set up communication pairs, submit transaction operations, and query responses. Further programming abstractions are supported based on the programming models. For example, unified remote procedure call (URPC) can implement remote procedure calls between any UBPU based on memory objects.

- **UBFM:** Oversees a UB domain. It manages compute, communication, and interconnect resources in the UB domain. Depending on the scale of a UB domain, multiple UBFM instances can be deployed to collaboratively manage resources within the domain.
- **UMMU:** Handles memory address mapping and permission verification during memory access. UMMUs facilitate memory resource sharing among UBPU and ensure that all memory access is properly authorized.
- **Security:** UB allows resource pooling and orchestration within a data center and supports multi-tenancy. To ensure secure resource access, UB employs advanced features such as device identity authentication, resource access isolation, access control, transmission confidentiality and integrity protection (CIP) for data paths, and extension of cross-device trusted execution environments (TEEs) to enable trusted resource sharing and orchestration, protecting data both in use and in transit.

View More

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